VisioSphere — Strong Review of Related Literature

All four detection modules: **fall**, **lying down**, **inactivity**, **agitation** — plus validation of the "no universal formula" claim. Peer-reviewed only · 2020 or later · every source opened on the publisher page and confirmed for venue, year and reported numbers. Preprints, theses, conference abstracts and papers without metrics excluded. 38 numbered sources. Rev. 10 September 2026.

Inclusion bar. A source was accepted only if all four held: (1) peer-reviewed journal or major conference proceedings; (2) 2020 or later; (3) topically direct to a measurement VisioSphere actually performs, or to the generalisation claim; (4) reports real numbers — accuracy, AUC, sensitivity, false-alarm rate, clinical outcome statistics, or measured threshold values in real units. Every source listed here is one you can cite.

1. FALL — the velocity discriminator

Fall and lying-down share one state machine in VisioSphere and are separated only by the **speed** of the transition (torso angular velocity ≥ 45 deg/s = FALL). These sources validate that architecture: static posture alone cannot tell a fall from a deliberate lie-down, so a velocity term is structurally required.

The core architectural argument: posture alone is not enough

[1] E. Tîrziu, A.-M. Vasilevschi, A. Alexandru, and E. Tudora, "Enhanced Fall Detection Using YOLOv7-W6-Pose for Real-Time Elderly Monitoring," *Future Internet*, vol. 16, no. 12, art. 472, 2024.
https://doi.org/10.3390/fi16120472

Supports: Your single best citation for the fall-vs-lying-down split. A YOLO-pose pipeline that states your argument verbatim: "we have differentiated between a fall and someone who is just lying down… When a fall occurs, there is a rapid change in body position… If the vertical speed exceeds a specific threshold, it indicates a fall, as opposed to someone slowly lying down." Same model family as yours, same decision.

Quotable figures: Torso-to-legs angle threshold **45 degrees**. Le2i, 130 videos: accuracy **95.7%**, precision 94.7%, recall 96.4%, specificity 95%, F1 95.5% (TP 54, TN 58, FP 3, FN 2). It confirms the speed threshold exists but does not publish its value.

[2] W. Chen, Z. Jiang, H. Guo, and X. Ni, "Fall Detection Based on Key Points of Human-Skeleton Using OpenPose," *Symmetry*, vol. 12, no. 5, art. 744, 2020.
https://doi.org/10.3390/sym12050744

Supports: The closest published **structural** match to your fall gate: a three-condition AND over (i) descent speed of the hip-joint centre, (ii) body-centreline-to-ground angle, (iii) bounding-rectangle width/height ratio. Your gate is the same three families of measurement combined the same way.

Quotable figures: Descent threshold stated verbatim as **0.009 m/s** for the hip-joint centre, computed "every five adjacent frames, with a time interval of 0.25 s"; angle condition θ **< 45°**; ratio $P \geq 1$. Sensitivity **98.3%**, specificity 95%, accuracy 97% on 100 actions, 10 subjects. **Caution:** quote 0.009 m/s exactly as printed — it is implausibly small for a true descent rate and reads as a normalised per-interval quantity. Be ready to say so if asked.

[3] D. Kibet, M. S. So, H. Kang, Y. Han, and J. Shin, "Sudden Fall Detection of Human Body Using Transformer Model," *Sensors*, vol. 24, no. 24, art. 8051, 2024.
https://doi.org/10.3390/s24248051

Supports: Vision-only, and the label set proves your point directly: **"falling," "lie," "lie down," "sit down," "sleeping," "standing," "stand up" are separate classes** sharing postural end-states, and only dynamics separate them. Features are keypoint positions *plus per-keypoint velocities* from inter-frame displacement.

Quotable figures: Accuracy **97.6%** over 600 clips, 7 participants, 100 per class. Velocities are min-max normalised, so no thresholds in physical units are published.

Is velocity from 2D pose keypoints trustworthy at all?

[4] R. Michaels, T. V. Barreira, S. N. Robinovitch, J. J. Sosnoff, and Y. Moon, "Estimating hip impact velocity and acceleration from video-captured falls using a pose estimation algorithm," *Scientific Reports*, vol. 15, art. 1558, 2025.
https://doi.org/10.1038/s41598-025-85934-y

Supports: The metrological defence of your whole velocity term. Validates landmark velocity recovered from 2D video pose against VICON motion capture (100 Hz) and an IMU (125 Hz). It also tells you what *not* to claim — acceleration from video is unreliable, which is precisely why your gate uses angular velocity rather than impact acceleration. That is a defensible design decision you can now cite.

Quotable figures: OpenPose on 30 fps / 1920×1080. Mean hip impact velocity 2.32 m/s. **Velocity: MAE 0.17 ± 0.13 m/s, MAPE 7.28%, bias −1.27%, r = 0.83** ("good accuracy with minimal bias"). **Acceleration: MAE 0.57 ± 0.53 g, MAPE 26.3%, r = 0.67** — "poor accuracy," underestimating impacts above 3.0 g.

Angular velocity named as the discriminant; temporal validation

[5] H. Shi, X. Wang, and J. Shi, "Fall Detection Algorithm Using Enhanced HRNet Combined with YOLO," *Sensors*, vol. 25, no. 13, art. 4128, 2025.

https://doi.org/10.3390/s25134128

Supports: A 2025 paper that independently arrives at **angular velocity between the trunk and the ground** as a fall criterion — the same feature as yours — alongside centre-of-mass decline speed and the height-to-width ratio. Also serves §2: it publishes actual aspect-ratio and tilt cut points, which most papers do not.

Quotable figures: Fall requires aspect **A < 1** AND descent **> 350 px/s** AND **trunk tilt > 75°**; normal walking **1.5 ≤ A < 3**. Fall accuracy **> 95%** at 18.1 fps; keypoint AP 73.49% (COCO). **Cite it for the feature, not the number** — it publishes no numeric angular-velocity threshold.

[6] V. Kong, S. Soeng, M. Thon, W.-S. Cho, A. Nayyar, and T.-K. Kim, "PIFR: A novel approach for analyzing pose angle-based human activity to automate fall detection in videos," *PLOS ONE*, vol. 20, no. 6, art. e0325253, 2025.
https://doi.org/10.1371/journal.pone.0325253

Supports: Nine pose-angle features (torso angle from vertical, hip and shoulder orientation, knee-joint angles, nose-to-ankle orientation) plus an explicit **temporal-validation stage over sequential frames** keyed on the "speed of posture change." Quotable against any panelist who asks why you do not decide per frame: "reliance on single-frame analysis increases false positive rates, as transient postures are often misinterpreted as falls." Also serves §2 for the angle family.

Quotable figures: URFD + Multiple Cameras Fall Dataset: precision **88.8%**, recall **94.1%**, F1 91.4%, specificity 95.6%, **false-positive rate 4.4%**, 23 FPS.

Holding identity through occlusion

[7] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," in *Computer Vision – ECCV 2022*, LNCS vol. 13682, Cham: Springer, 2022, pp. 1–21.
https://doi.org/10.1007/978-3-031-20047-2_1

Supports: Your tracker, cited at its **peer-reviewed venue** — use this, never the arXiv preprint. Its mechanism is exactly the occlusion argument your per-track state machine depends on: low-score boxes, which typically arise from occlusion, are matched by similarity to tracklets rather than being "simply thrown away," which otherwise causes "missed objects and fragmented trajectories."

Quotable figures: MOT17 test: **MOTA 80.3, IDF1 77.3, HOTA 63.1**; state of the art on MOT20, HiEve and BDD100K.

2. LYING DOWN — the LYING_EVIDENCE positive-evidence gate

Backing the four conditions of the new gate: lower-body visibility, the torso-normalised vertical stack ratio, the bounding-box aspect limits, and the hip-extension / torso-angle alternatives. References **[5]** and **[6]** above also apply here.

Scale-invariant body-segment ratios — backs your stack ratio

[8] K. Han, Q. Yang, and Z. Huang, "A Two-Stage Fall Recognition Algorithm Based on Human Posture Features," *Sensors*, vol. 20, no. 23, art. 6966, 2020.
https://doi.org/10.3390/s20236966

Supports: The closest published analogue to your vertical stack ratio. Computes a **spine ratio** = spine vector length ÷ waist-width vector length from OpenPose 2D keypoints — the same construction as yours: divide one body measurement by another measured in the same frame, so camera distance cancels.

Quotable figures: Decision boundary **RB = 1.0**; ADL frames cluster above **1.2**, fall frames below **1.2** — **strikingly close to your LYING_STACK_MAX of 1.20**. Angle thresholds 30° / 50°. Accuracy **97.34%**, precision 98.50%, recall 97.33%, F1 97.91%.

[9] B. M. V. Guerra, S. Ramat, G. Beltrami, and M. Schmid, "Automatic Pose Recognition for Monitoring Dangerous Situations in Ambient-Assisted Living," *Frontiers in Bioengineering and Biotechnology*, vol. 8, art. 415, 2020.
https://doi.org/10.3389/fbioe.2020.00415

Supports: The clearest published *statement* of why you normalise. Standing / sitting / lying / dangerous-sitting from joint angles plus vertical joint coordinates **explicitly normalised by subject height**, "to eliminate subject-specific variability." Also documents your occlusion problem: sitting frames are noisy precisely because joints are hidden by other body parts.

Quotable figures: 602,530 frames, 12 subjects, held-out-subject test. Mean accuracy **83.9% ± 0.73%**; **lying 94.3%** (AUC 98.5), standing 93.8%, sitting 86.9%. Sitting↔dangerous-sitting confusion 6.01% / 2.91%.

Keypoint-confidence gating and temporal confirmation

[10] H. Rehouma and M. Boukadoum, "Fall Detection by Deep Learning-Based Bimodal Movement and Pose Sensing with Late Fusion," *Sensors*, vol. 25, no. 19, art. 6035, 2025.
https://doi.org/10.3390/s25196035

Supports: Your best citation for **MANDATORY 1** (lower-body visibility at conf ≥ 0.45) and for temporal confirmation. Excludes landmarks below a visibility threshold *before* any geometric reasoning, and requires a run of consecutive frames before alarming — the same architecture as LYING_EVIDENCE_MIN_FRAC / MIN_FRAMES. Also your fusion ablation in §5. **State openly:** its second channel is a waist IMU, not vision, so it supports the architecture, not a pure-vision claim.

Quotable figures: Visibility cut **0.50**; aspect < 1 upright, > 1 prone; **30 consecutive frames (1 s)** to confirm. F1 **97.3%**, FPR **3.6%** vs **8.9%** vision-only and **11.3%** IMU-only — a 57% cut over vision alone. False positives from "abrupt sitting" explicitly suppressed. 16 participants, LOSO, < 5 lux.

The bending-over false positive is a known, quantified problem

[11] L. Qi and H. Sun, "Multistage fall detection framework via 3D pose sequences and TCN integration," *Scientific Reports*, vol. 15, art. 27832, 2025.
https://doi.org/10.1038/s41598-025-11325-y

Supports: Evidence that your false alert of 09-09 — a resident bent over a corner table measuring torso 97°, aspect 0.74, extent 1.46 — is a **recognised field-wide failure mode**, not a defect unique to VisioSphere. The paper deliberately stress-tests against sitting down, squatting and shoe-removal.

Quotable figures: NTU RGB+D: **890 fall samples vs 2,800 deliberately similar non-fall samples**. Accuracy 99.87%, macro-F1 97.68%, **FP and FN rates as low as 0.13%**. Best with 16 joints.

[12] A. Bustamante, L. M. Belmonte, R. Morales, A. Pereira, and A. Fernández-Caballero, "Bridging the Appearance Domain Gap in Elderly Posture Recognition with YOLOv9," *Applied Sciences*, vol. 14, no. 21, art. 9695, 2024.
https://doi.org/10.3390/app14219695

Supports: Problem framing and baseline: standing / sitting / lying discrimination in **real elderly subjects** from a single camera is a well-posed, high-accuracy task. **Note honestly:** bounding-box only — no keypoints, angles or ratios. It supports the task, not your gates.

Quotable figures: Best config accuracy **98.41%**, F1 0.97; lying precision 0.93 / recall 0.95. Useful warning for §5: virtual-to-real transfer collapses to **70.64% accuracy, F1 0.25**.

3. INACTIVITY — the long lie

This is the module with the **strongest clinical literature of the four**, and the one where the evidence is medical rather than computational. Your duration threshold is not an arbitrary engineering choice — the one-hour mark is a published clinical convention with outcome data behind it. Lead with this section if a panelist asks which detection is best justified.

What a "long lie" is, and why one hour

[13] J. Kubitza, I. T. Schneider, and B. Reuschenbach, "Concept of the term long lie: a scoping review," *European Review of Aging and Physical Activity*, vol. 20, art. 16, 2023.
https://doi.org/10.1186/s11556-023-00326-3

Supports: The definitional citation for your entire inactivity threshold. Establishes that the field defines a long lie as remaining on the floor for **one hour or longer** — and, usefully for your honesty, notes the term "is determined neither by a fall nor by a fixed time period." That second clause protects you: the literature itself concedes the threshold is a convention.

Quotable figures: 22 studies, **6,311 individuals**; 2,805 unable to get up; 1,121 affected by a long lie. Mean lying times across cohorts: **1.9 days**, **18.8 hours**, **~2 hours**. "On average, older adults wait **nine to twenty minutes** until they get help." Mean hospitalisation 27 / 14 / 2 days. At 12 months, **29–77%** move into long-term care. Ambulance re-contact **78.8% with long lie vs 60.6% without**.

[14] J. Kubitza, M. Haas, L. Keppeler, and B. Reuschenbach, "Therapy options for those affected by a long lie after a fall: a scoping review," *BMC Geriatrics*, vol. 22, art. 582, 2022.
https://doi.org/10.1186/s12877-022-03258-2

Supports: Supplies the **prevalence** half of the argument: how often this actually happens. Full text verified.

Quotable figures: 19 studies, **1,276 subjects**, 5 countries. "**One in eight older individuals who fall report lying on the floor for more than an hour.**" More than half of older people living alone cannot get up or get help independently. Half of those over 65 who fall are found on the floor by rescue services. Among persons requiring care, **70%** of those who fell already had problems getting up.

What a long lie does to the body — the hard-outcome source

[15] C. Hüser, M. Hackl, V. Suárez, I. Gräff, M. Bernhard, V. Burst, and C. Adler, "Liegetrauma: retrospektive Analyse einer Patientenkohorte aus einer universitären Notaufnahme" [Long lie trauma: retrospective analysis of a patient cohort presenting to a university hospital emergency department], *Medizinische Klinik – Intensivmedizin und Notfallmedizin*, vol. 118, no. 3, pp. 220–227, 2023.
https://doi.org/10.1007/s00063-022-00912-w

Supports: The paper that turns your inactivity alert from a convenience feature into a safety one. An emergency-department cohort of long-lie patients with complication and mortality rates. If a panelist asks why inactivity deserves an alert at all, this is the answer. Peer-reviewed original research; German-language, online-first 2022.

Quotable figures: n = 50, median age 76. Documented lying time **median 13.5 h (range 9–96 h)**. Complications: **dehydration 66%**, infections 52% (pneumonia 69% of those), **severe rhabdomyolysis 21% at admission, 26% peak**, acute kidney injury 20%,

severe hypothermia (< 32 °C) 20%. **ICU admission 69%. In-hospital mortality 50%.** Only 48% of survivors went home.

That automated detection actually reduces floor time

[16] E. Bayen, S. Nickels, G. Xiong, J. Jacquemot, R. Subramaniam, P. Agrawal, R. Hemraj, A. Bayen, B. L. Miller, and G. Netscher, "Reduction of time on the ground related to real-time video detection of falls in memory care facilities: Observational study," *Journal of Medical Internet Research*, vol. 23, no. 6, art. e17551, 2021.
https://doi.org/10.2196/17551

Supports: The single most on-point source in this entire document. A camera-based fall and floor-time system deployed across six memory-care facilities, measuring exactly the outcome your inactivity detector claims to improve. It also gives you the unwitnessed-fall figure. **Distinguish it honestly at defence:** SafelyYou Guardian is **human-in-the-loop** — staff review the video — whereas VisioSphere alerts automatically. That difference is your contribution, and naming it first is stronger than being asked.

Quotable figures: 436 falls in 66 residents (mean age 87), 10 months, 6 facilities. Mean Time On Ground **43.2** → **22.9** → **13.6 min**; Time Until Assistance **35.3** → **15.9** → **7.0 min**. Baseline→real-time: TOG −29.6 min (95% CI 20.3–38.9, p = .043); TUA −28.3 min (p = .004). **Fallers with TOG > 1 hour: 31% (8/26) at baseline** → **zero events.** And: a staff member was in the room during only **17.4% (44/253)** of falls — i.e. **~83% were unwitnessed**.

Automated immobility detection — computational precedent

[17] R. Analia, A. Forster, S.-Q. Xie, and Z. Zhang, "Privacy-preserving approach for early detection of long-lie incidents: A pilot study with healthy subjects," *Sensors*, vol. 25, no. 12, art. 3836, 2025.
https://doi.org/10.3390/s25123836

Supports: The closest published analogue to your inactivity architecture: a vision-based fall detector followed by **rule-based logic on posture stability and duration** to declare a long lie. Direct precedent for "detect the posture, then time it."

Quotable figures: Fall-vs-ADL: accuracy **0.98**, precision 0.98, recall 0.98, F1 0.98. 320 fall + 150 ADL sequences; long-lie decision over a **15-minute continuous window**. **Concede at defence:** 10 healthy young subjects (mean age 29.4), not residents, and no separate metrics for the long-lie stage itself.

[18] I. Susnea, E. Pecheanu, A. Cocu, A. Istrate, C. Anghel, and P. Iacobescu, "Non-intrusive monitoring and detection of mobility loss in older adults using binary sensors," *Sensors*, vol. 25, no. 9, art. 2755, 2025.
https://doi.org/10.3390/s25092755

Supports: Independent corroboration that **"inactivity longer than one hour" is the accepted operational unit** in ambient elderly monitoring — a second discipline arriving at your threshold.

Quotable figures: CASAS Aruba (220 days) and HH120 (64 days). Detection rate **~77–81%** for point anomalies at a **false-positive rate of 12–16%**. Enumerated **60** and **36** inactivity periods exceeding one hour respectively. Also detected simulated mobility decline of 1% per day.

Why staff reporting alone is not enough

[19] N. Albasha, C. Curtin, R. McCullagh, N. Cornally, and S. Timmons, "Staff's insights into fall prevention solutions in long-term care facilities: a cross-sectional study," *BMC Geriatrics*, vol. 23, art. 738, 2023.
https://doi.org/10.1186/s12877-023-04435-7

Supports: The "why not just rely on staff" answer. **Be precise at defence:** this is self-reported staff practice, not an audit of unwitnessed falls. The 29% figure is the quotable line.

Quotable figures: 155 staff, 13 of 14 LTC sites. Post-fall practice (n = 131): checked for injuries 55.0%, called for help 42.7%, **completed an incident report only 29.0% (38/131)**, updated the care plan 9.9%. Asked what would prevent falls, staff nominated **monitoring/surveillance 31.31%** and alarm systems 19.38%.

Alarm fatigue — why your escalation ladder and circuit breaker are safety features

[20] S. Chen, H. Zhi, S. Wu, R. Jiang, H. Zhang, H. Zhang, and H. Zhang, "Determinants of medical equipment alarm fatigue in practicing nurses: A systematic review," *SAGE Open Nursing*, vol. 9, art. 23779608231207227, 2023.
https://doi.org/10.1177/23779608231207227

Supports: The headline false-alarm burden numbers. Pair with [21] and with [35] in §5 to argue that rate limiting (ALERT_MAX_PER_HOUR = 30) is a clinical safeguard, not a shortcut.

Quotable figures: 14 studies, **2,848 nurses**. Pooled alarm-fatigue score 21.76 (95% CI 20.27–23.25). "**72%–99% of clinical alarms may be false alarms.**" One included study: **8,336 alarms in 4 weeks, of which only 333 (4%) were actionable**. Nurses may face alarms **up to 1,000 times in a single shift**.

[21] H. Ruppel, M. Dougherty, C. P. Bonafide, and K. B. Lasater, "Alarm burden and the nursing care environment: a 213-hospital cross-sectional study," *BMJ Open Quality*, vol. 12, no. 4, art. e002342, 2023.
https://doi.org/10.1136/bmjoq-2023-002342

Supports: The best evidence that alarm burden **measurably degrades staff response** — the mechanism, not just the count. This is the justification for an escalation ladder rather than a flat alarm.

Quotable figures: 3,986 nurses, 213 hospitals. 83% felt overwhelmed by alarm volume; 76% reported **delaying response**; 55% encountered “a patient needed urgent attention, but no one responded”; 46% had work interrupted by non-actionable alarms. In hospitals rated poor on safety, **72%** experienced unresponded urgent alarms vs **38%** in good ones ($p < 0.001$). Overwhelmed nurses reported **3.9 missed care activities** vs 2.5 ($p < 0.001$).

4. AGITATION — reversal counting and the CV rhythm test

Extrema / reversal detection in a pose trajectory

[22] H. L. Cornman, J. Stenum, and R. T. Roemmich, “Video-based quantification of human movement frequency using pose estimation: A pilot study,” *PLOS ONE*, vol. 16, no. 12, art. e0261450, 2021.
https://doi.org/10.1371/journal.pone.0261450

Supports: Validates your **reversal-counting front end** directly: detecting local minima and maxima in a pose-estimated limb trajectory accurately identifies discrete movement events and derives movement frequency.

Quotable figures: Event-time correlation with manual annotation **r > 0.99** across all tasks at 1–4 Hz; mean frequency error ≤ **0.03 Hz**; **993 vs 1002** finger taps detected (pose vs manual); typical event-time error 1–2 frames.

[23] F. L. Marshall, S. Zhang, and B. Scotney, “Automatic Assessment of the Type and Intensity of Agitated Hand Movements,” *Journal of Healthcare Informatics Research*, vol. 6, no. 4, pp. 401–422, 2022.
https://doi.org/10.1007/s41666-022-00120-3

Supports: The closest published system to your agitation module: agitated hand movement quantified from RGB video by tracking a hand centroid as a 1-D trajectory, **peak detection at local extrema**, intensity expressed as **repetitions per minute**. Backs both your reversal counting and your rate gate (AGITATION_OSC_MIN_RATE). **This is the real source of the 30–90 rpm figure** — see the correction notice in §7.

Quotable figures: Metronome-paced intensities **30, 50, 70, 90 rpm**; 20 participants, 400 × 30 s sequences. Type accuracy 80.0%, **intensity accuracy 92.5%**, both correct 75.25%. Type window **6 s**; **intensity window 12 s with 5 s overlap**. **Concede:** 20 healthy actors simulating agitation, not people with dementia.

Coefficient of variation as a rhythmicity index

[24] A. Numata, Y. Terao, K. Sugawara, Y. Ugawa, and T. Furubayashi, “Differences in the movement phase condition and sensory inputs on temporal synchronization and continuation during bilateral foot-tapping tasks,” *Frontiers in Human Neuroscience*, vol. 19, art. 1518230, 2025.
https://doi.org/10.3389/fnhum.2025.1518230

Supports: Your methodological precedent: **CV of inter-tap intervals as the primary index of rhythmic timing stability** in repetitive human movement — the same statistic you apply to inter-reversal intervals.

Quotable figures: CV of ITI, SC-tap: unilateral **0.055 ± 0.027**, bilateral in-phase 0.051, antiphase 0.047 ($p = 0.011$). SCA-tap: 0.063 / 0.061 / 0.057 ($p = 0.009$). **Use as the lower anchor** — metronome-paced lab taps are the regularity ceiling, which is why your rhythmic band sits above these values.

[25] S. Boripuntakul, T. Kamnardsiri, S. R. Lord, S. Maiarin, P. Worakul, and S. Sungkarat, “Gait variability during abrupt slow and fast speed transitions in older adults with mild cognitive impairment,” *PLOS ONE*, vol. 17, no. 10, art. e0276658, 2022.
https://doi.org/10.1371/journal.pone.0276658

Supports: Supplies the **formal definition** you can cite for the statistic — CV = (SD / mean) × 100 applied to inter-event intervals of a repetitive human movement — and shows it discriminates a cognitively impaired group from controls.

Quotable figures: Step-time CV significantly higher in MCI vs control on the slow-to-fast transition, **p = 0.001**.

[26] N. A. Timmermans, R. Terranova, D. C. Soriano, H. Cagnan, Y. P. Raykov, I. G. Bucur, B. R. Bloem, R. C. Helmich, and L. J. W. Evers, “A generalizable and open-source algorithm for real-life monitoring of tremor in Parkinson’s disease,” *npj Parkinson’s Disease*, vol. 11, art. 205, 2025.
https://doi.org/10.1038/s41531-025-01056-2

Supports: Large-scale validation of the principle behind your rhythm test: detect a repetitive motor behaviour using features that **explicitly capture periodicity**, gated by a frequency-band threshold. Also a model for external validation in your Methodology.

Quotable figures: 24 + 24 training, **517 PD + 50 controls external validation**. Sensitivity 0.61 (SD 0.20), specificity 0.97 (SD 0.05); **peak-frequency gate 3–7 Hz**; test–retest ICC 0.98.

Repetitive behaviour and agitation from video — feasibility in the real setting

[27] T. Barami, L. Manelis-Baram, H. Kaiser, M. Ilan, A. Slobodkin, O. Hadashi, D. Hadad, D. Waissengreen, T. Nitzan, I. Menashe *et al.*, “Automated Analysis of Stereotypical Movements in Videos of Children With Autism Spectrum Disorder,” *JAMA Network Open*, vol. 7, no. 9, art. e2432851, 2024.
https://doi.org/10.1001/jamanetworkopen.2024.32851

Supports: Automated video detection and **counting** of repetitive stereotyped movements at clinical scale. Cite for “repetitive movements are reliably extractable from video” — not for the CV method, which this paper does not use.

Quotable figures: 220 children, 319 assessments, **580 hours of video**. Accuracy **92.53%** (95% CI 81.09–95.10).

[28] S. S. Khan, P. K. Mishra, N. Javed, B. Ye, K. Newman, A. Mihailidis, and A. Iaboni, "Unsupervised Deep Learning to Detect Agitation From Videos in People With Dementia," *IEEE Access*, vol. 10, pp. 10349–10358, 2022.
https://doi.org/10.1109/ACCESS.2022.3143990

Supports: The most directly comparable deployment in the literature: **existing ceiling cameras in a long-term care dementia unit** detecting clinically significant agitation. Same setting, same constraints, same class-imbalance problem as VisioSphere.

Quotable figures: Trained on ~24 h of normal activity, tested on 11 h with annotated agitation events. **AUC(ROC) = 0.754.** The modest AUC is useful for calibrating panel expectations about what is realistically achievable in this setting.

[29] P. K. Mishra, A. Iaboni, B. Ye, K. Newman, A. Mihailidis, and S. S. Khan, "Privacy-protecting behaviours of risk detection in people with dementia using videos," *BioMedical Engineering OnLine*, vol. 22, art. 4, 2023.
https://doi.org/10.1186/s12938-023-01065-3

Supports: Your **ethics-chapter citation**: anonymising video to skeleton keypoints costs essentially nothing in detection performance versus raw RGB. This is the published justification for a pose-only pipeline in a care facility.

Quotable figures: Toronto Rehabilitation Institute dementia unit; 15 wall-mounted cameras, 352×240 @ 30 fps; ~600 days of footage. RGB AUC(ROC) **0.822**; **skeleton 0.812**; segmentation mask 0.823. AUC(PR) 0.124–0.134 — honest evidence of the imbalance.

[30] A. Badawi, S. Elmoghazy, S. Choudhury, S. Elgazzar, K. Elgazzar, and A. M. Burhan, "Multimodal Detection of Agitation in People With Dementia in Clinical Settings: Observational Pilot Study," *JMIR Aging*, vol. 8, art. e68156, 2025.
https://doi.org/10.2196/68156

Supports: Current state of the art for camera-based agitation detection in a real inpatient dementia unit, faces anonymised. Shows the video channel alone is viable.

Quotable figures: 10 participants, single site. **Video: GRU 95% accuracy, AUC 0.99**; wearable Extra Trees up to 99% personalised. Agitation predicted ≥ 6 min in advance. **Present the AUC honestly — n = 10 pilot.**

[31] M. Al-Hammadi, H. Fleyeh, and I. Thomas, "Gait and movement analysis for discrimination between people with dementia and healthy control persons based on pose estimation and machine learning," *Journal of Alzheimer's Disease*, vol. 111, no. 1, pp. 222–239, 2026.
https://doi.org/10.1177/13872877261430001

Supports: Validates that **markerless 2D pose keypoints from ordinary RGB video** carry reliable motor signal in a dementia population. Cite for pose-estimation validity in your population — it is not an agitation study.

Quotable figures: 64 people with dementia vs 67 controls; YOLOv8, **17 keypoints, mean keypoint visibility 0.77**. Best SVM **87% ± 5.1 accuracy**, 86.6% recall.

5. Validating the claim: "there is no single universal formula"

The evidence base for your defence argument — that direct calculation-level RRLs are scarce *because the field has no consensus formula*, and that the strongest recent work combines complementary measurements rather than perfecting one.

The same model collapses on a different dataset

[32] V. Fula and P. Moreno, "Wrist-Based Fall Detection: Towards Generalization across Datasets," *Sensors*, vol. 24, no. 5, art. 1679, 2024.
https://doi.org/10.3390/s24051679

Supports: The cleanest demonstration that a near-perfect published accuracy is a property of the dataset, not of the formula.

Quotable figures: Within-dataset AUC-ROC **99.25%**, recall 95.16%. Cross-dataset: UMAFall→UPFall AUC **68.34%**, recall **45.16%**; WEDAFall→UPFall AUC **65.23%**. Merging all three datasets restores recall to 90.57% — **only more diverse data recovers performance, not a better formula**.

No consensus formula; reported accuracies are not comparable

[33] M. Ishaq, D. C. Guastella, G. Sutera, and G. Muscato, "A Systematic Review of Fall Detection and Prediction Technologies for Older Adults: An Analysis of Sensor Modalities and Computational Models," *Applied Sciences*, vol. 16, no. 4, art. 1929, 2026.
https://doi.org/10.3390/app16041929

Supports: The strongest single citation for your claim. The authors state a quantitative meta-analysis "was not feasible" because of "significant heterogeneity in sensor modalities, experimental protocols and performance metrics reported," and that "models that perform well on one dataset may not generalize to others."

Quotable figures: 130 studies. Reported accuracy spans **79.6–99.98%** (vision 79.6–99.98%, wearable 90.3–99.9%). **128 of 130 studies (98.5%) validated on simulated falls; only 2 tested real-world unanticipated falls.**

[34] P. Gorce and J. Jacquier-Bret, "Fall Detection in Elderly People: A Systematic Review of Ambient Assisted Living and Smart Home-Related Technology Performance," *Sensors*, vol. 25, no. 21, art. 6540, 2025.

https://doi.org/10.3390/s25216540

Supports: PRISMA systematic review, PROSPERO-registered, carrying **both halves of your claim at once**: measured heterogeneity, and measured superiority of multimodal approaches.

Quotable figures: 80 studies, **426 accuracy values pooled from 59 studies**. **63.5% used custom private datasets**, against 33 different open datasets. Wearable 91.1% / non-wearable 92.6% / **hybrid multimodal 94.6% accuracy, 97.6% sensitivity, 98.2% F1**; hybrid beat wearable-only at **p < 0.05**.

Complementary measurements beat any single one

Reference **[10]** is your fusion ablation: combined FPR **3.6%** against **8.9%** vision-only and **11.3%** IMU-only — a 57–67% cut in false positives from adding one complementary channel, with each component measured separately. Reference **[2]** makes the same point structurally: three conditions ANDed, none sufficient alone.

Real deployments perform nothing like lab benchmarks

[35] R. G. Curtis, C. Maher, K. A. Dankiw, B. Singh, and S. Bogomolova, "In-Home Evaluation of the Neo Care Artificial Intelligence Sound-Based Fall Detection System," *Future Internet*, vol. 16, no. 6, art. 197, 2024.
https://doi.org/10.3390/fi16060197

Supports: A **commercial single-modality** detector measured in real homes — what happens when one measurement carries the whole decision. It justifies your escalation ladder and circuit breaker as much as it supports the claim.

Quotable figures: 18 households, mean age 75.3, 12 weeks. Sensitivity **75.00%** but specificity **6.80%**. **4,939 activations, of which 4,930 were false** (9 true) — roughly **one false alarm every 1.25 hours**. Variability attributed partly to the individual home's layout.

Agitation specifically has no agreed measurement

[36] B. Wong, P. Wu, Z. Ismail, J. Watt, and Z. Goodarzi, "Detecting agitation and aggression in persons living with dementia: a systematic review of diagnostic accuracy," *BMC Geriatrics*, vol. 24, no. 1, art. 559, 2024.
https://doi.org/10.1186/s12877-024-05143-6

Supports: Even the **human** instruments disagree. If trained clinicians using validated scales cannot converge on what agitation is, a demand that your detector match a single universal formula is not a coherent standard. Quotable: "Until 2015, there lacked a consensus-based definition of agitation, and consequently a reference standard diagnosis."

Quotable figures: 36 studies. Specificity for the same construct ranges **28.6% to 98%** across instruments (NBRS 28.6–85.0%; PAS 57.1%; NPI-Spanish 97.8%). Meta-analysis impossible.

[37] M. Mbogori-Kairichi, A. Malioukis, R. S. Snead, J. Marczika, R. Ambalavanan, and G. Towett, "Digital biomarkers for early agitation detection in dementia: a scoping review of emerging wearable and smart technologies for personalized care," *Frontiers in Neurology*, vol. 17, art. 1683517, 2026.
https://doi.org/10.3389/fneur.2026.1683517

Supports: Shows the pattern in one table: high single-modality numbers measure narrow, easier problems; honest multimodal systems report far more sober figures. Quotable: "heterogeneity in how agitation was defined and measured across studies also constrains comparability."

Quotable figures: 13 studies from 798 screened. Single-modality **91.69–98.6%** task-specific accuracy, versus multimodal **median AUC 0.87, sensitivity 71.32%, specificity 75.28%**.

[38] N. Farina, L. Smith, M. Rajalingam, and S. Banerjee, "Implementation Outcomes for Agitation Detection Technologies in People with Dementia: A Systematic Review," *Geriatrics*, vol. 10, no. 3, art. 70, 2025.
https://doi.org/10.3390/geriatrics10030070

Supports: The single best justification for your project existing. It does not validate detector performance — it validates the gap. Do not cite it for accuracy numbers; it reports none.

Quotable figures: 1,697 records screened, 19 included. **Median sample size 10.** 63% multimodal; **only 1 study (5%) was camera-based**; 26% in residential care. Key finding: "a general absence of primary data related to implementation outcomes."

6. Carried forward — supporting, secondary

Source	Role and caution	Module
Zeng *et al.*, "SmoothNet: A Plug-and-Play Network for Refining Human Poses in Videos," *ECCV*, 2022.	Temporal filtering of keypoint trajectories — backs your 3-sample median filter before the rhythm test.	Agitation
Beddiar, Oussalah, and Nini, "Fall detection using body geometry and human pose estimation in video sequences," *JVCIR*, vol. 82, art. 103407, 2021.	Body-geometry ratios from pose — general support for the ratio family.	Fall / Lying

Source	Role and caution	Module
Inturi *et al.*, "A Novel Vision-Based Fall Detection Scheme Using Keypoints of Human Skeleton with LSTM," *Sensors*, vol. 23, no. 14, art. 6283, 2023.	Keypoint-based fall detection with temporal modelling.	Fall
Dentamaro, Impedovo, and Pirlo, "Fall Detection by Human Pose Estimation and Kinematic Theory," *ICPR*, pp. 2328–2335, 2021.	Angular kinematics for fall vs non-fall — secondary support for the angular-velocity split.	Fall
Gaya-Morey, Manresa-Yee, and Buades-Rubio, "Deep learning for computer vision based activity recognition and fall detection of the elderly: a systematic review," *Applied Intelligence*, vol. 54, no. 19, pp. 8982–9007, 2024.	Systematic review of vision-based elderly fall detection — survey-level support for the heterogeneity argument.	Claim
Nagaj *et al.*, *Smart Innovation, Systems and Technologies*, pp. 215–226, 2025.	Recent pose-based fall work.	Fall
Gattani *et al.*, *Ageing Research Reviews*, vol. 113, art. 102948, 2025; Raza *et al.*, *Engineering Applications of AI*, vol. 143, art. 109809, 2025; Nabizade *et al.*, *Sensors*, vol. 26, no. 16, art. 5069, 2026.	Survey-level breadth for the heterogeneity argument.	Claim

7. What the literature does NOT support — read before the defence

Four of your constants have **no citable 2020+ threshold behind them**. I searched specifically for each and found nothing meeting the bar. Claiming otherwise is the fastest way to lose the panel. Present each as your own empirically tuned parameter, cite the *method* paper for the measurement, and put the tuning evidence in Methodology.

Constant	What is missing	What to say instead
Fall: torso angular velocity ≥ **45 deg/s**	**No 2020+ peer-reviewed vision paper publishes an angular-velocity threshold in deg/s.** [5] names trunk-to-ground angular velocity as a discriminant but gives no value; [6] leaves "speed of posture change" configurable. The classic hard number (47.3 °/s, Bourke & Lyons, *Med. Eng. Phys.*, 2008) fails your bar twice — pre-2020 *and* gyroscope-based.	Cite [1] and [3] for "speed separates falls from lie-downs," [5] and [6] for angular velocity as an accepted feature, and [4] for velocity-from-pose being metrologically sound. State 45 deg/s as your calibration. Note in passing that Bourke's 47.3 °/s is remarkably close — convergent, not derived.
Fall: pixel-aspect correction $k = (W_{src}/H_{src})/(W_{inf}/H_{inf})$	**Zero peer-reviewed sources.** Searches returned only ImageMagick threads, Cognex product docs and patents. The academic camera-calibration literature covers *lens* distortion, not anisotropic resize invalidating an atan2 on pixel coordinates.	**Present this as an engineering contribution, not a cited step.** Every vision fall-detection paper in this list computes body angles directly on pixel coordinates with no aspect correction at all. That is a real gap you are closing — claim it.
Lying: **LYING_HIP_EXTEND_MIN = 140°** and **LYING_STACK_MAX = 1.20**	No published hip-flexion degree cut-point separating supine/prone from seated in 2020+ pose-based classification. No paper publishes a torso-normalised stack ratio with your 2.5–3.0 / 1.4–2.0 / 1.1–1.8 / 0–1.0 bands.	Cite [6] (hip and knee angles *are* discriminative) and [8] (angle thresholds of 30°/50° are publishable and effective). For the ratio, [8]'s spine ratio has boundary 1.0 with an empirical split at **1.2** against your 1.20 — lead with that parallel, it is genuinely strong.
Agitation: **AGITATION_OSC_MAX_CV = 0.55 / 0.40**	No paper reports or justifies a numeric CV cut-off separating rhythmic from random human movement. Also **no 2020+ peer-reviewed source states and measures CV ≈ 1 for a Poisson event train in a behavioural context** — that is pre-2020 statistical theory.	Cite [24] and [25] for CV-of-intervals as the accepted rhythmicity statistic, and attribute the Poisson property to standard theory (Cox & Lewis, *The Statistical Analysis of Series of Events*, 1966). Your *mc_agitation.py* Monte Carlo table is the evidence for the cut-off — **lead with it**.
Inactivity: duration threshold	This one is the exception — **it is well supported.** The one-hour convention is published in [13], the prevalence in [14], the outcomes in [15], the effect of reducing floor time in [16], and independent corroboration in [18]. Searched and not found: an *adjusted* odds or hazard ratio for time-on-floor published 2020+; all such estimates are pre-2020.	Present Hüser's **50% in-hospital mortality** as a raw rate, not an adjusted effect estimate. Bayen's **31%** → **0** for falls exceeding one hour is your strongest single number in the whole document.